Sri Chaitanya IIT Academy.,India.

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A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT **Paper -2(Adv-2024-P2-Model)** **Date: 19-04-2026**

Time: 02.30Pm to 05.30Pm **GTA-25** **Max. Marks: 180**

KEY SHEET

MATHEMATICS

1	B	2	D	3	C	4	C	5	AD	6	CD
7	AC	8	27	9	7	10	882	11	92	12	756
13	58	14	12	15	2	16	0.5	17	1		

PHYSICS

18	B	19	C	20	D	21	D	22	ABD	23	ACD
24	AD	25	4	26	4	27	3	28	7	29	16
30	9	31	40	32	4.50	33	5	34	2.5		

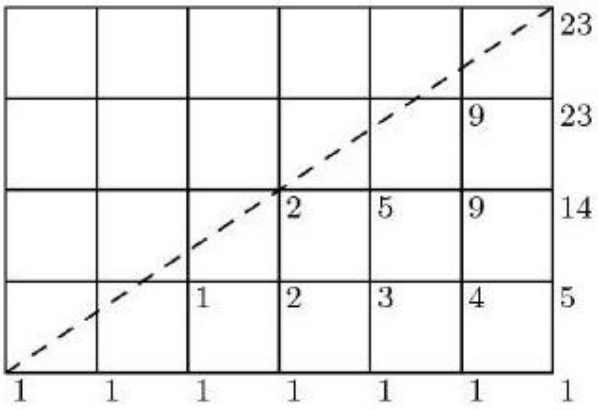
CHEMISTRY

35	B	36	C	37	D	38	C	39	BCD	40	ABCD
41	CD	42	300	43	80	44	28	45	8	46	5
47	18	48	20	49	18.80 to 18.84	50	8	51	6.66 to 6.67		

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SOLUTIONS
MATHEMATICS

1. We graph the 10 shots on a grid. Suppose that a made shot is represented by a step of $(0,1)$, and a missed shot is represented by $(1,0)$. Then the basketball player's shots can be represented by the number of paths from $(0,0)$ to $(6,4)$ that always stay below the line $y=\frac{2x}{3}$. We can find the number of such paths using a Pascal's Triangle type method below, computing the number of paths to each point that only move right and up.



Therefore, there are 23 ways to shoot 4 makes and 6 misses under the given conditions. The probability of each possible sequence occurring is $(.4)^4(.6)^6$. Hence the desired probability is $\frac{23\cdot 2^4\cdot 3^6}{5^{10}}$, and the answer is $(23+2+3+5)(4+6+10)=660$.

2. Hyperbola: $7x^2+8xy-y^2-4=0$

Asymptotes: $7x^2+8xy-y^2=0$

$C:(0,0)$

(hand : $11x+3y=14$

Let asymptotes ly $y=mx$, where $m=m_1,m_2$.

$$m_1+m_2=8, m_1m_2=-7$$

Equation of circumircle of $\Delta CEF=x^2+y^2-ax-by+c=0$ passes through $C(0,0)$

For E and F_1 salve $y=mx$ and $11x+3y=14$

$$11x+3mx=14\Rightarrow x=\frac{14}{11+3m}, y=\frac{14m}{11+3m}$$

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$\therefore \varepsilon, F\left(\frac{14}{11+3m}, \frac{14m}{11+3m}\right)$ for m_1, m_2.

Now F and F lices on circle.

$$\left(\frac{14}{11+3m}\right)^2+\left(\frac{14m}{11+3m}\right)^2-\frac{14a}{11+3m}-\frac{14bm}{11+3m}=0$$

$$(3b-14)m^2+(3a+11b)m+11a-14=0$$

None $m_1+m_2=8, m_1m_2=-7$

$$\therefore 3a+35b=112$$

$$11a+21b=112$$

$$\therefore\ a=\frac{112}{23}, b=\frac{64}{23}, c=0$$

3. $f(x)$ is a linear polynomial

$\therefore f(x)=2x+1$

4. $f(x)=1,\ g(x)=2^{10}\cdot x^{10}$

5. $p(x)=(x-\alpha)\left(x^2-2\beta x+\beta^2+\gamma^2\right)$

$p'(x)=3x^2-2(\alpha+2\beta)x+\beta^2+\gamma^2+2\alpha\beta$

$\Delta=(\alpha-\beta)^2-3\gamma^2$

6. $f(x)=\left(-n^3\right)\left(\left|2\tan^{-1}\left(e^{|x|}-e\right)-\frac{1}{n^3}\right|-2\left|\tan^{-1}\left(e^{|x|}-e\right)\right|\right)$

Let $g(x)=e^{|x|}-e$

$$=\lim_{n\to\infty}\frac{\left(-n^3\right)\left(\left(2\tan^{-1}g(x)-\frac{1}{n^3}\right)^2-4\left(\tan^{-1}g(x)\right)^2\right)}{\left|2\tan^{-1}g(x)-\frac{1}{n^3}\right|+2\left|\tan^{-1}g(x)\right|}$$

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$$= \lim_{n\to\infty} \frac{\left(-n^3\right)\left(\frac{-4\tan^{-1}g(x)}{n^3}+\frac{1}{n^6}\right)}{\left|2\tan^{-1}g(x)-\frac{1}{n^3}\right|+2\left|\tan^{-1}g(x)\right|} = \frac{\tan^{-1}g(x)}{\left|\tan^{-1}g(x)\right|} x \neq \pm 1$$

$$f(x)=\begin{cases}\dfrac{\tan^{-1}g(x)}{\left|\tan^{-1}g(x)\right|} & x\neq\pm1\\ -1 & x=\pm1\end{cases}$$

(A) $f(x)$ is discontinuous $x=\pm1$

(B) $|f(x)|$ is continuous function

7. $f(x)=-\frac{1}{4}(x^3+3x^2-5x-3)$

From that $b+d=0$ and $a+c=1$

$$\int_{-1}^{1}(bx^2+cx+d)dx=1$$

$$\Rightarrow 2\int_{0}^{1}(bx^2+d)dx=1$$

$$\Rightarrow b=\frac{-3}{4}, d=\frac{3}{4}$$

now

$$I=\int_{-1}^{\frac{1}{2}} f''(x^2)dx=\int_{-1}^{\frac{1}{2}}(6ax-\frac{3}{2})^2=\frac{27}{2}((a+\frac{1}{4})^2+\frac{3}{16})$$

$$I_{\min}=\frac{81}{32} \text{ at } a=\frac{-1}{4}\Rightarrow c=1-a=\frac{5}{4}$$

hence $f(x)=-\frac{1}{4}(x^3+3x^2-5x-3)$

8. A, B, C lie on a circle of radian $\frac{\sqrt{3}}{2}$

ΔABC is equilateral and its plane is $\frac{1}{2}$ units from $(0, 0, 0)$

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$$\therefore V = 6\left(\frac{\sqrt{3}}{4}\cdot\frac{3}{4}\cdot\frac{1}{2}\right) = \frac{9\sqrt{3}}{16}$$

9. $$y_n = \left(1+\frac{1}{n^2}\right)\left(1+\frac{4}{n^2}\right)^2 - \left(1+\frac{n^2}{n^2}\right)^n$$

$$L = \lim_{n\to\infty}\left[\left(1+\frac{1}{n^2}\right)\left(1+\frac{4}{n^2}\right)^2 - \left(1+\frac{n^2}{n^2}\right)^n \frac{-4}{n^2}\right.$$

$$\log L = \lim_{n\to\infty}\frac{-4}{n}\Sigma_{r=1}^{n}\frac{r}{n}\log\left(1+\frac{r^2}{n^2}\right)$$

$$\log L = -4\int_0^1 x\log\left(1+x^2\right)dx$$

$$L = \frac{e^2}{16}$$

10. $$\prod_{i=0}^{r-1}(n+r-i)(n-i)$$

$$\left(\prod_{i=0}^{r-1}(n+r-i)\right)\left(\prod_{i=0}^{r-1}(n-i)\right) = \frac{(n+r)!}{(n-r)!}$$

$$S_n = \Sigma_{r=1}^{n-1}\left(\frac{n^2-2r^2+n-3r-1}{(r+1)^2(r!)^2}\right)\frac{(n+r)!}{(n-r)!}$$

$$S_n = \Sigma_{r=1}^{n-1}\frac{\left(n^2-2r^2+n-3r-1\right)}{(r+1)^2}c_r^n c_r$$

Let $A_r = {}^{n+r}C_r\,{}^nC_r$

$$S_n = \Sigma_{r=1}^{n-1}\frac{\left(n^2-2r^2+n-3r-1\right)}{(r+1)^2}A_r$$

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$$A_{r+1}=\frac{(n+r+1)!}{(n-r-1)!((r+1)!)^2}=\frac{(n+r+1)(n-r)}{(r+1)^2}A_r$$

$$A_{r+1}-A_r=A_r\left[\frac{(n+r+1)(n-r)}{(r+1)^2}-1\right]$$

$$A_{r+1}-A_r=A_r\left[\frac{n^2-2r^2+n-3r-1}{(r+1)^2}\right]$$

$$S_n=\sum_{r=1}^{n-1}(A_{r+1}-A_r)$$

$$S_n=A_n-A_1={}^{2n}C_n-n(n+1)$$

11. $$\sum_{x=2}^{44}2\sin x\sin 1\left[1+\sec(x-1)\sec(x+1)\right]$$

$$\Rightarrow\sum_{x=2}^{44}\left[\cos(x-1)-\cos(x+1)\right]+\sum_{x=2}^{44}\left[x_c(x+1)-\sec(x-1)\right]$$

$$-\frac{\sin^2 1}{\cos 1}-\frac{\sin^2 2}{\cos 2}+\frac{\sin^2 44}{\cos 44}+\frac{\sin^2 45}{\cos 45}$$

12. Just to visualize solution 1. If we list all possible $(x,f(x))$, from $1,2,3,4,5$ to $1,2,3,4,5$ in a specific order, we get $5\cdot 5=25$ different $(x,f(x))$'s. Namely:

$$\begin{matrix}(1,1)(1,2)(1,3)(1,4)(1,5)\\(2,1)(2,2)(2,3)(2,4)(2,5)\\(3,1)(3,2)(3,3)(3,4)(3,5)\\(4,1)(4,2)(4,3)(4,4)(4,5)\\(5,1)(5,2)(5,3)(5,4)(5,5)\end{matrix}$$

To list them in this specific order makes it a lot easier to solve this problem. We notice, In order to solve this problem at least one pair of (x,x) where $x\in 1,2,3,4,5$ must exist. In this case I rather "go backwards". First fixing 5 pairs (x,x), (the diagonal of our table) and map them to the other fitting pairs $(x,f(x))$. You can do this in $\frac{5!}{5!}=1$ way. Then fixing 4 pairs

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(x,x) (The diagonal minus 1) and map them to the other fitting pairs $(x,f(x))$. You can do this in $4\cdot\frac{5!}{4!}=20$ ways. Then fixing 3 pairs (x,x) (The diagonal minus 2) and map them to the other fitting pairs $(x,f(x))$. You can do this in $\frac{(5\cdot4\cdot3\cdot6\cdot3)}{3!2!}+\frac{(5\cdot4\cdot3\cdot6\cdot1)}{3!}=150$ ways. Fixing 2 pairs (x,x) (the diagonal minus 3) and map them to the other fitting pairs $(x,f(x))$. You can do this in

$$\frac{(5\cdot4\cdot6\cdot4\cdot2)}{2!3!}+\frac{(5\cdot4\cdot6\cdot4\cdot2)}{2!2!}+\frac{(5\cdot4\cdot6\cdot2\cdot1)}{2!2!}=380 \text{ ways.}$$

Lastly, fixing 1 pair (x,x) (the diagonal minus 4) and map them to the other fitting pairs. You can do this in $\frac{5!}{4!}+4\cdot\frac{5!}{3!}+5!=205$ ways.

So $1+20+150+380+205=756$

13. First, we use the fact that $f(f(x))=x$ for all x in the domain. Substituting the function definition, we have $\dfrac{a\dfrac{ax+b}{cx+d}+b}{c\dfrac{ax+b}{cx+d}+d}=x$, which reduces to

$$\frac{\left(a^2+bc\right)x+b(a+d)}{c(a+d)x+\left(bc+d^2\right)}=\frac{px+q}{rx+s}=x.$$

In order for this fraction to reduce to x, we must have $q=r=0$ and $p=s\neq0$.

From $c(a+d)=b(a+d)=0$, we get $a=-d$ or $b=c=0$. The second cannot be true, since we are given that a,b,c,d are nonzero.

This means $a=-d$, so $f(x)=\dfrac{ax+b}{cx-a}$.

The only value that is not in the range of this function is $\frac{a}{c}$. To find $\frac{a}{c}$, we use the two values of the function given to us. We get $2(97)a+b=97^2c$ and $2(19)a+b=19^2c$. Subtracting the second equation from the first will eliminate b, and this results in $2(97-19)a=\left(97^2-19^2\right)c$, so

$$\frac{a}{c}=\frac{(97-19)(97+19)}{2(97-19)}=58.$$

Alternatively, we could have found out that $a=-d$ by using the fact that

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$f\left(f\left(-b/a\right)\right)=-b/a$.

14. $\left(x^2+x+1\right)P\left(x-1\right)=\left(x^2-x+1\right)P\left(x\right)$

Since, x^2+x+1 and x^2-x+1 have no factor

Let $p\left(x\right)=\left(x^2+x+1\right)\phi\left(x\right)$

From (1), $\phi\left(x-1\right)=\phi\left(x\right)$

$\Rightarrow \phi\left(x\right)=\phi\left(x+1\right)\Rightarrow$ periodic

A polynomial is periodic if it is constant

$\Rightarrow P\left(x\right)=k\left(x^2+x+1\right), P\left(1\right)=3$

$\Rightarrow P\left(x\right)=x^2+x+1$

$$\text{Now}, \int_0^1 \tan^{-1}\left(\frac{2x}{1+P\left(x^2\right)}\right)dx=\int_0^1 \tan^{-1}\left(\frac{2x}{1+x^4+x^2+1}\right)dx$$

$$=\int_0^1\left\{\tan^{-1}\left(x^2+x+1\right)-\tan^{-1}\left(x^2-x+1\right)\right\}dx=\int_0^1\left\{\tan^{-1}\left(\frac{1}{1+x^2-x}\right)-\tan^{-1}\left(\frac{1}{1+x^2+x}\right)\right\}dx$$

$$=\int_0^1\left\{\tan^{-1}x-\tan^{-1}\left(x-1\right)-\tan^{-1}\left(1+x\right)+\tan^{-1}x\right\}dx$$

$$\int_0^1 \tan^{-1}\left(\frac{2x}{1+P\left(x^2\right)}\right)dx+\int_0^1 \tan^{-1}\left(1+x\right)dx=3\left[\frac{\pi}{4}-\frac{1}{2}\ln 2\right]=\frac{3}{4}\left[\pi-\ln 4\right]\Rightarrow k=12$$

15. $I_n=2^n\int_0^1\left(1+x^4\right)^n dx$

$$\Rightarrow I_n=2^n\left[x\cdot\left(1+x^4\right)^n\Big|_0^1-\int_0^1 x\cdot n\cdot 4x^3\left(1+x^4\right)^{n-1}dx\right]$$

$$\Rightarrow I_n=2^n\cdot 2^n-2^n\cdot 4n\int_0^1\left(1+x^4-1\right)\left(1+x^4\right)^{n-1}dx$$

$$\Rightarrow I_n=4^n-4n\cdot 2^n\int_0^1\left\{\left(1+x^4\right)^n-\left(1+x^4\right)^{n-1}\right\}dx$$

$$\Rightarrow I_n=4^n-4n\cdot I_n+8n\cdot I_{n-1}$$

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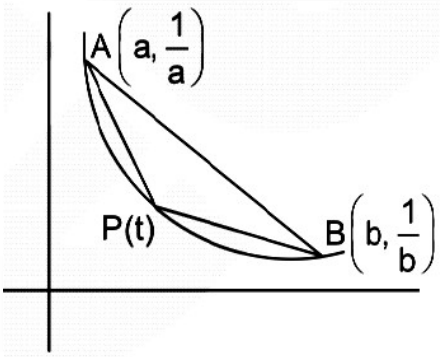
$\Rightarrow (1+4n)I_n = 4^n + 8nI_{n-1}$

By comparison, $4^n = 8n$

$\Rightarrow n = 2$

16. $A(t) = \dfrac{(a-b)(a-t)(b-t)}{2abt}$

$A'(t) = \dfrac{(a-b)\left(t^2-ab\right)}{2abt^2}$

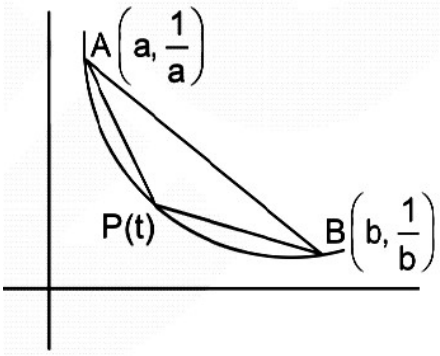


So, $t = \sqrt{ab}$ is point of maxima moreover

$$\int_a^{\sqrt{ab}}\left(-\frac{1}{at}x+\frac{1}{a}+\frac{1}{t}-\frac{1}{x}\right)dx = \int_{\sqrt{ab}}^{b}\left(-\frac{1}{bt}x+\frac{1}{b}+\frac{1}{t}-\frac{1}{x}\right)dx$$

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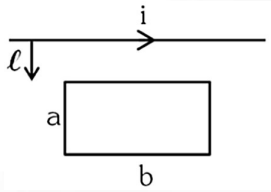
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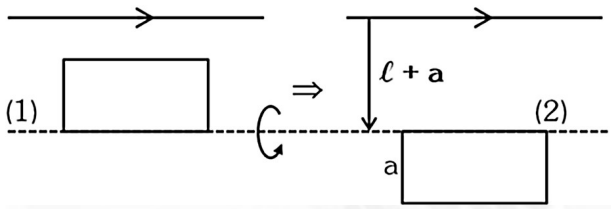
PHYSICS

18. Sol.



$$\phi_i=\int_{\ell}^{\ell+a}\frac{\mu_0 i}{2\pi x}b\,dx=\frac{\mu_0 ib}{2\pi}\ln\left(\frac{\ell+a}{\ell}\right)$$

When rotated about farthest side :



$$\phi_f=\frac{\mu_0 ib}{2\pi}\ln\left(\frac{\ell+2a}{\ell+a}\right)$$

$$q_1=\left|\frac{\Delta\phi}{R}\right|=\frac{\mu_0 ib}{2\pi}\ln\left(\frac{\ell+a}{\ell}\times\frac{\ell+2a}{\ell+a}\right)$$

Instead of rotating if shifted to position 2 :

$$\phi_f=\frac{\mu_0 ib}{2\pi}\ln\left(\frac{\ell+2a}{\ell+a}\right)$$

$$q_2=\left|\frac{\Delta\phi}{R}\right|=\frac{\mu_0 ib}{2\pi R}\ln\left(\frac{\ell+a}{\ell}\times\frac{(\ell+a)}{(\ell+2a)}\right)$$

When carried away to large distance $\phi_f=0$

$$q_3=\left|\frac{\Delta\phi}{R}\right|=\frac{\mu_0 ib}{2\pi R}\ln\left(\frac{\ell+a}{\ell}\right)$$

From eq. (1) & (2) :-

$$q_3=\frac{q_1+q_2}{2}$$

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19. $\frac{\lambda}{2} = 45\,\text{cm} = 0.45\,\text{cm}$

$f = \frac{u}{\lambda} = \frac{160}{2 \times 0.45} = 177.7\,Hz$

20. The light is travelling along negative x-axis; therefore, line of sight is positive x-axis.

Before, $\tan\theta_b = \frac{E_y}{E_z} = \frac{-6}{8} = \frac{-3}{4} \Rightarrow \theta_b = -37^\circ$

After, $\tan\theta_a = \frac{E_y}{E_z} = \frac{3}{3\sqrt{3}} = \frac{1}{\sqrt{3}} \Rightarrow \theta_a = 30^\circ$

21. $f_{beat} = 10Hz$

For carrier oscillation $f = \frac{20}{0.02} = 1000Hz$

22. For newton's ring, radius

$r = \sqrt{n\lambda R}$

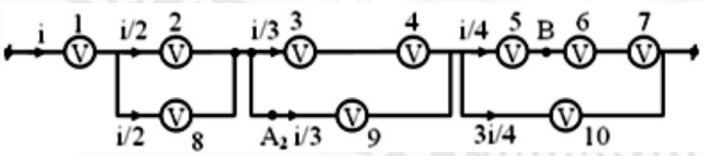
$r_{\max} = \sqrt{n_{\max}\lambda R} = \sqrt{50 \times 600 \times 10^{-9} \times 8}$

$r_{\max} = \sqrt{50 \times 4.8 \times 10^{-6}} \approx 1.54\,\text{cm}$

- At center there is a dark spot which is due to phase difference of π on reflection so $t = 0$
- $\frac{1}{f} = (n_{rel} - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$

$f = \frac{R}{(n_{rel} - 1)} = \frac{8}{0.5} = 16\,\text{m}$

23.



$V_1 = iR = 12\text{ V}$

(A) $V_2 = \frac{1}{2}R = 6V$

(B) $V_4 = \frac{1}{3}R = 4V$

(C) $V_6 = 1/4R = 3V$

(D) $V_A - \frac{2i}{3}R - i/4R = V_B$

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$$V_A - V_B = \frac{2i}{3}R + \frac{i}{4}R = 8 + 3 = 11\ V$$

24. Lamp stays on if $\frac{v}{4} < V_1 < v/2$, so capacitor voltage oscillates between $V/4$ and $V/2$.

Capacitor charges through R_1 from supply V when lamp is off. When lamp turns on capacitor discharges quickly through lamp as $R_- \ll R_1$.

(A). Slow charging (time constant R_1C) and rapid discharging (time constant $R-C$).

$$V_C = V + (V_1 - V)e^{-t/R_1C}$$

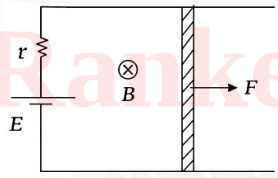
(D) Flash frequency $\rightarrow$ charging frequency (as discharging is relatively very quick)

$$f = \frac{1}{t_{\text{charge}}}\ t_{\text{charge}} = R_lC\ln\left(\frac{v - v_1}{v - v_2}\right)$$

$$= 5000 \times 100 \times 10^{-6}\ln\left(\frac{v - v/4}{v - v/2}\right)$$

$$\approx 4.95\ \text{Hz}$$

25.



- EMF induced in rod $\epsilon_0 = BV_0l\cos(\omega t)$
- current $i = \frac{E - BVl}{r}$

$$F_{\text{net}} = F + Bil = m\frac{dv}{dt}$$

$$F + \frac{Bl}{r}(E - Bvl) = m\frac{dv}{dt}$$

$$F = -mv_0\omega\sin\omega t - \frac{BlE}{r} + \frac{B^2l^2}{r}v_0\cos\omega t$$

$$P = \vec{F}\cdot\vec{v} = -m\omega v_0^2\sin\omega t\cos\omega t - \frac{BlE}{r}v_0\cos\omega t + \frac{B^2l^2v_0^2}{r}\cos^2\omega t$$

$$\langle P\rangle - \frac{1}{2}m\omega v_0^2\langle\sin 2\omega t\rangle - \frac{BlEv_0}{r}\cos\omega t + \frac{B^2l^2v_0^2}{r}\langle\cos^2\omega t\rangle$$

$$\langle P\rangle = 4W$$

26. Network $= 2P_0V_0 - P_0V_0 - \frac{\pi P_0V_0}{4} = 0.22P_0V_0$

Heat is the in $1 \rightarrow 2$ only (ΔT is +ve)

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$Q_{1\to 2} = 4.5P_0V_0 + 1.22P_0V_0 = 5.72P_0V_0$

$\eta = \dfrac{0.22P_0V_0}{5.72P_0V_0} \approx 0.04$ or $\eta = 4\%$

27. no. of wavelengths in A $n_A = \dfrac{t}{u_A}.f$

no. of wavelengths in B $n_B = \dfrac{t/3}{u_B}.f$

no. of wavelengths in C $n_C = \dfrac{2t/3}{u_C}.f$

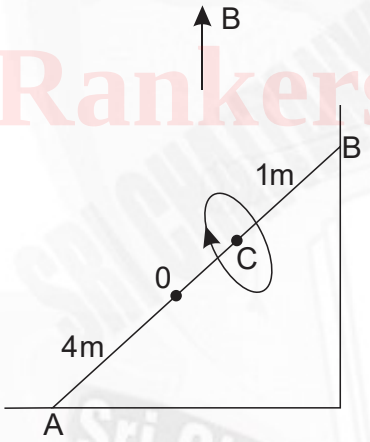
Now, $\dfrac{tf}{v_A} = \dfrac{t/3\, f}{u_B} + \dfrac{\left(2t/3\right)f}{u_C}$

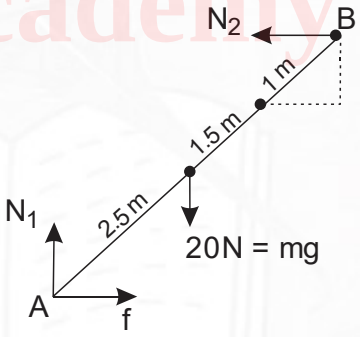
$\dfrac{n_A}{C} = \dfrac{n_B}{3C} + \dfrac{2n_C}{3C}$

$\Rightarrow 1.5 = \dfrac{n_B}{3} + \dfrac{2}{3}.(1.6)$

$\Rightarrow n_B = 4.5 - 3.2 = 1.3$ $\quad\therefore\quad \boxed{x = 3}$

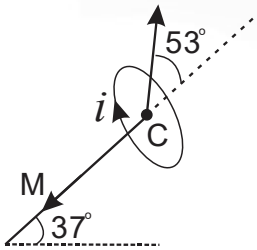
28.





$M = iA = i(\pi r^2) = (1)\pi\dfrac{1}{\pi} = 1\ \text{Am}^2$

$\sum F_{100p} = 0$ {$\because$ B = uniform}



$\tau_c = M\,B\sin(53^\circ) = (1)(20)\left(\dfrac{4}{5}\right) = 16\,\text{Nm}\,(\text{CW})$

Equilibrium $\Rightarrow N_1 = 20\,N,\ N_2 = f$

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Also, $\sum P_c = 0 \Rightarrow$

$$\Rightarrow N_2(i\sin 37^\circ)(CCW) + mg(1.5\cos 37^\circ)(CCW) + 16(CW) + mg = N_1(4\cos 37^\circ)(CW) + f(4\sin 37^\circ)(CCW) = 0$$

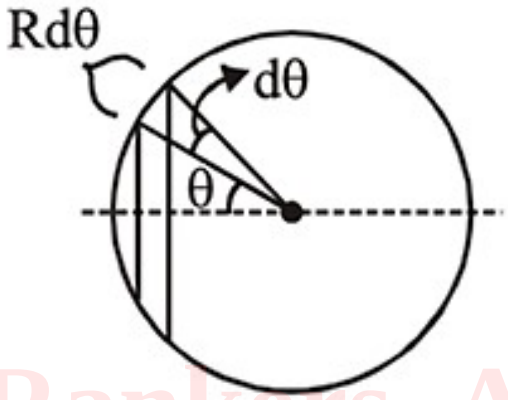
$$f\left(\frac{3}{5}\right) + (20)(1.5)\left(\frac{4}{5}\right) + \frac{12f}{5} = 16 + 80\left(\frac{4}{5}\right)$$

$$3f + 120 + 12f = 80 + 320 \Rightarrow 15f = 280 \Rightarrow 15\mu(mg) = 280 \Rightarrow 15\mu = 14$$

$$\Rightarrow \mu = \frac{2x}{15} = \frac{14}{15}$$

$$\Rightarrow \boxed{x = 7}$$

29.



Let us first calculate magnetic field B at centre of bigger ring due to its rotation about axis xx'.

$$dB = \frac{\mu_0 (di)(R\sin\theta)^2}{2R^3},\ di = \frac{dq}{T} = \frac{(dq)\omega}{2\pi} = \frac{Q\omega}{4\pi^2}d\theta$$

$$\left\{dq = \frac{Q}{2\pi R}(Rd\theta) = \left(\frac{Q}{2\pi}\right)d\theta\right\}$$

$$\Rightarrow dB = \frac{\mu_0}{2R}\left(\frac{Q\omega}{4\pi^2}\right)\sin^2\theta d\theta \Rightarrow B = \frac{\mu_0 Q\omega}{8\pi^2 R}\int_0^\pi \sin^2\theta d\theta$$

$$\Rightarrow B = \frac{\mu_0 Q\omega}{8\pi^2 R}\left(\frac{\pi}{2}\right) = \frac{\mu_0 Q\omega}{16\pi^2 R} = \frac{\mu_0 Q(\alpha t)}{16\pi R}$$

$$\varepsilon_1 = A\frac{dB}{dt} = \frac{(\pi a^2)(\mu_0 Q\alpha)}{16\pi R} = \frac{\mu_0 Q\alpha a^2}{16R}$$

$$\Rightarrow \varepsilon_1 = \frac{\mu_0\left(\frac{1600}{\mu_0}\right)(8)(0.01)}{16(1)} = 8V$$

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$\cdot current = \frac{\varepsilon}{r} = 8A$

30. Least count of screw gauge $= \frac{0.5}{50} = 0.1$ mm

n VSD $= 9$MSD $= 9$ mm

Least count of caliper $= 1$MSD $- 1$SSD

$$1 - \frac{9}{n} = 0.1$$

$$0.9 = \frac{9}{n} \therefore n = 10$$

So, $(n-1) = 9$

31. For trajectory of rocket

$$a = \frac{16C + 2C}{2} = 9C$$

$$E = -\frac{GMm}{18C}$$

So, $\frac{1}{2}mV_A^2 - \frac{GMm}{26} = -\frac{GMm}{18C}$

$$V_A = \sqrt{\frac{8}{9}}\sqrt{\frac{GMs}{C}} = \frac{2\sqrt{2}}{3}.30\sqrt{2} = 40\,\text{km/hr}$$

32. $\frac{T_{\text{ellipse}}}{T_A} = \left(\frac{q_{\text{ellipse}}}{rA}\right)^{3/2}$

$$T_{\text{ellipse}} = \frac{2\sqrt{2}}{3} \cdot \frac{27}{2\sqrt{2}} = 9 \text{ years}$$

$$T_{AB} = \frac{T_{\text{ellipse}}}{2} = 4.5 \text{ years.}$$

33. Focal length of polished lens

$$\frac{1}{f} = \frac{2}{f_l} + \frac{1}{f_m} = 2\left(\frac{3}{2} - 1\right)\left(\frac{1}{40} - \frac{1}{-40}\right) + \frac{1}{20}$$

$f = 10$ cm (converging mirror)

By mirror formula

$$\frac{1}{v} + \frac{1}{-a} = -\frac{1}{10} \therefore v = \frac{10a}{(10-a)} - (1)$$

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$$m = -\frac{u}{u} = 2 = \frac{10a/(10-a)}{-a}$$

$$\therefore a = 5 \text{ cm}$$

34. Since, distance of object O from plane mirror M is (a + b), therefore, it forms a virtual image at a distance (a + b) behind itself or a distance 2 (a + b) from the object O. Since, there is no parallax between the images formed by the silvered lens L and plane mirror M, therefore, two images are formed at the same point. Hence, distance of Image from lens L is 2 (a + b) - a = (a + 2b) behind lens. Since, length of Image formed by L is twice the length of image formed by the mirror M and length of image formed by a plane mirror is always equal to length of the object, therefore, modulus of transvers magnification produced by the lens L is equal to 2.

(a+2b) = 2a or b = a/2 ……(1)

The slivered lens L may be assumed as a combination of an equi-convex lens and a concave mirror placed in contact with each other co-axially as shown in Fig.



For the lens

$R_1 = +r, R_2 = -r, \mu = 1.5$

Its focal length f_1 is given by

$$\frac{1}{f_1} = (\mu - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) \text{ for } f_1 = 40 \text{ cm}$$

For concave mirror, R = -r = - 40 cm

Its focal length, $f_m = \frac{R}{2} = -20\text{cm}$

The combination L behaves like a mirror whose equivalent focal length F is given by

$$\frac{1}{F} = \frac{1}{f_m} - \frac{2}{f_1} \text{ or } F = -10\text{cm}$$

Hence, for the combination,

$u = -a, v = +2a, F = -10\text{cm}$

Using mirror formula, $\frac{1}{v} + \frac{1}{u} = \frac{1}{F}$

a = 5 cm

Substituting a = 5 cm in equation (1) b = 2.5 cm

CHEMISTRY

35. Since alkali metals have lowest ionization energy, y should be the smallest number. Also because removing third electron from magnesium atom breaks its noble gas core, z should be the biggest number. Since $w < 2081 < x$; clearly $w < x$; so $y < w < x < z$ should be the correct order.

36. Molarity = 2M

= 2 moles NH_3 / L of solution

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$= 2\times17 = 34\text{g } NH_3 / \text{L of solution} \Rightarrow \frac{w}{V} = \frac{34}{1000}\times100 = \boxed{3\cdot4}$

$= 34\text{g } NH_3 / 1500\text{g of solvent} \Rightarrow \frac{w}{W} = \frac{34}{1534}\times100 = \frac{34}{15\cdot24}$

$= 2\,\text{moles}\,NH_3 / 1.5\,\text{kg of solvent} \Rightarrow \frac{v}{NH_3} = \frac{2}{2+\frac{1500}{18}} = \frac{3}{128}$

$= \frac{4}{3}\text{moles}\,NH_3 / \text{kg of solvent}$

$= \frac{4}{3}\text{m}$

37. $\frac{r_{H_2}}{r_{gas}} = \sqrt{\frac{M_{gas}}{M_{H_2}}} = 3\sqrt{3} = \sqrt{27}$

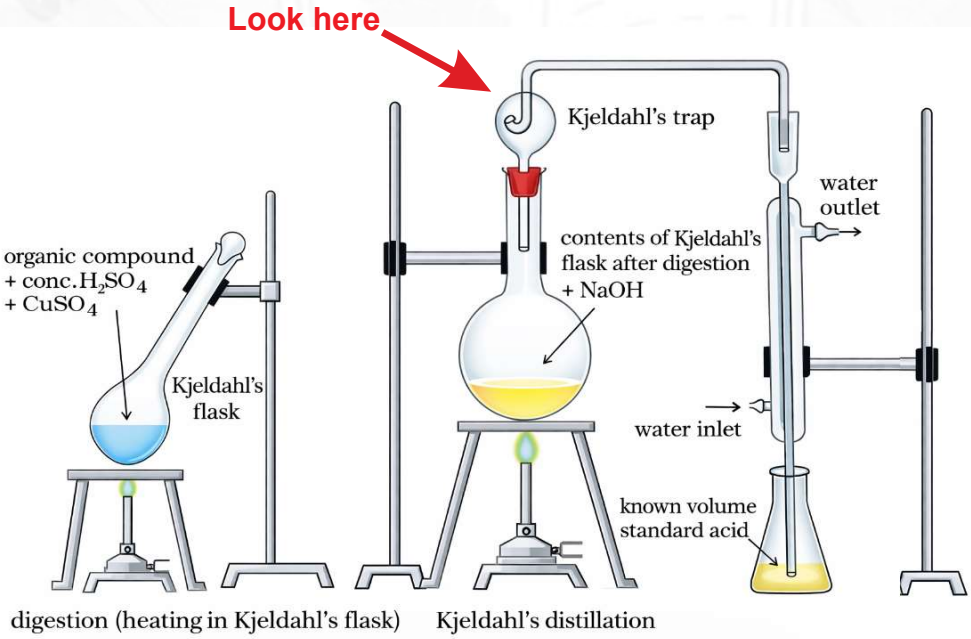
$\therefore \sqrt{27} = \sqrt{\frac{M_{gas}}{2}}$

$M_{gas} = 54\text{ g / mol}$

Gaseous hydrocarbon with mass 54 will be C_4H_6

Because d.o.u. = 2 the hydrocarbon may be an alkyne alkadiene or bicycloalkane but not a spiroalkane because even the simplest spiroalkane contains 5 carbon atoms.

38. Refer NCERT Class XI Unit 8 Fig. 8.16 and lab manual



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39. a & b)

Under the wire: The heavy weights apply high pressure to a small area of the ice. This lowers the melting point below $0^\circ C$. Even if the ice is at $0^\circ C$, it will melt into water because of this pressure.

Above the wire: As the wire moves down into the newly formed liquid, the water left behind is no longer under pressure. Since the surrounding ice is still at $0^\circ C$, this water refreezes into solid ice. As a result the wire will slowly pass through the ice block without splitting it into two.

c) Adding salt lowers the freezing point of ice which facilitates melting of ice and hence the wire moves down faster.

d) In a normal environment, the surrounding air (slightly above $0^\circ C$) acts as a medium that can provide a tiny amount of thermal energy through convection.

In a vacuum, you lose all heat transfer via convection. The system becomes purely dependent on the thermal conductivity of the wire itself and the ice. Without the surrounding air to nudge the temperature or provide a thermal buffer, the overall movement will be significantly slower because the heat exchange gets restricted.

40.

- A chain : 21 amino acids $\Rightarrow (21-1)=20$ peptide bonds

  B chain : 30 amino acids $\Rightarrow (30-1)=29$ peptide bonds

  Total peptide bonds $=49$

- As seen in the structure given, of the 3 disulphide bonds, **2 are interchain** and **1 is intrachain**
- A chain contains 1 glycine unit while B chain contains 3 glycine units.
- Aspartic acid (Asp), Methionine (Met) and Tryptophan (Trp) are not present in the structure shown.

41. Compound X is **NaOH** and Compound Y is $\mathbf{Na_2CO_3 \cdot 10\ H_2O}$

42. $\dfrac{40\times10^{-6}\times193\times60}{96500}$ Faradays

$=4\cdot8\times10^{-6}$ Faradays

$=4\cdot8\times10^{-6}$ Equivalents of Ag deposited

$=4\cdot8\times10^{-6}$ moles of Ag deposited

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$= 4\cdot8\times10^{-6}\times6\times10^{23}$ atoms of Ag deposited

$= 2\cdot88\times10^{18}\times1.5625\times10^{-16}\,\text{cm}^2$ area

$= 4.5\times10^{2}\,\text{cm}^2$ area

$= 450\ \text{cm}^2$ area (which is 60% of cathode area)

$\therefore$ unplated area $= 40\% = \dfrac{40}{60}\times450 = \mathbf{300}\,\text{cm}^2$

43. $r = \dfrac{-d[A]}{dt} = \dfrac{-1}{2}\dfrac{d[B]}{dt} = +\dfrac{1}{3}\dfrac{d[C]}{dt}$

$r_A = \dfrac{1}{2}\times r_B = \dfrac{1}{2}\times10^{-3} = 0.5\times10^{-3}\ \text{ms}^{-1}$

rate of heat released $= \left(0.5\times10^{-3}\times400\right) = 0.2\ \text{kJ}/l/s$

For temperature to remain constant, reaction (2) must absorb heat at the same rate as it is released

$\Delta H_E = +300\ \text{kJ/mole of}\ E$

rate of disappearance of E,

$= \dfrac{r_{\text{heat absorption}}}{\Delta H_E}$

$= \dfrac{0.2\,\text{kJ}/\ell/\text{s}}{300\ \text{kJ/mol}}$

$= \dfrac{1}{1500}\ \text{M/s}$

rate of appearance of F,

$r_F = 2\times r_E = 2\times\dfrac{1}{1500}$

$= \dfrac{1}{750}\ \text{Ms}^{-1}$

$= \dfrac{1}{750}\ \text{M}\times60\ \text{min}^{-1}$

$= 0.08\ \text{M/min}$

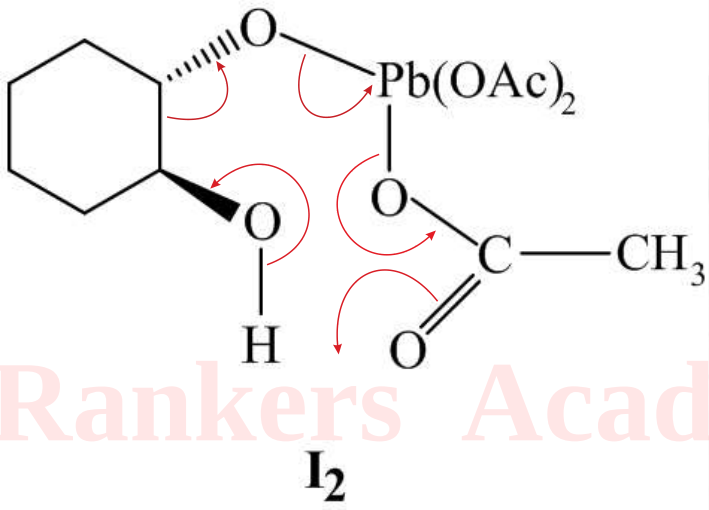
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$$= \mathbf{80} \times 10^{-3} \text{ M / min}$$

44. $2NH_3 + 3CuO \rightarrow 3Cu + 3H_2O + N_2$

Nitrogen is an inert gas with very low boiling point, so it does not condense in ice-bath. It continues through the tube & is collected by downward displacement of water.

45. 

46. 

$\mathbf{I_2}$

47. Inulin has one glucose unit at the bottom; and rest are $n+2$ fructose units.

total units $= 1 + (n+2) = n+3$

total no. of glycosidic bonds $= (n+3) - 1 \quad = (n+2)$

Each bond eliminates one H_2O molecule (18 g / mol)

mass before condensation $= 180(n+3)$

water Lost $= 18(n+2)$

final molar mass $= 3420$

$$180(n+3) - 18(n+2) = 3420$$

$$180n + 540 - 18n - 36 = 3420$$

$$162n = 2916$$

$$n = \mathbf{18}$$

48. $V_{drop} = \frac{4}{3}\pi r^3 = 0.036\pi \text{ cm}^3$

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$$m_{drop} = d \times v = \frac{100}{9\pi} \times 0.036\pi = 0.4g = 400mg$$

Area occupied by drop $= \pi r^2 = 16\pi \text{cm}^2$

Rate of absorption $= \frac{5}{4\pi} \times 16\pi = 20 \text{mg/min}$

Time taken $= \frac{400}{20} = \mathbf{20}$ minutes

49. Rate of adsorption = 20 mg/min

Rate of evaporation = $= \frac{5}{4} = 1.25$ mg/min

Rate of disappearance of drop = 21.25 mg/min

Time taken $= \frac{400}{21.25} = \mathbf{18.82}$ minutes

50. $\underset{(g)}{PCl_5} \rightleftharpoons \underset{(g)}{PCl_3} + \underset{(g)}{Cl_2}$

initially, $\underset{2}{} \quad \underset{2}{} \quad \underset{2}{}$

Total moles at $t = 0 = 6$ moles

$\because$ moles are equal, partial pressure of each gas

$$= \frac{1}{3} \times P_{\text{total}}$$

$$\therefore P_{PCl_5} = P_{PCl_3} = P_{Cl_2} = 1 \text{atm}$$

$$K_p = \frac{P_{PCl_5} \cdot P_{Cl_2}}{P_{PCl_5}} = 1$$

Now, Cl_2 is added at (g) constant pressure & temperature until volume becomes 2V.

$PV = nRT$

If $P, T =$ constant

$V \propto n$

if V becomes $2V$, n becomes $2n$

$\therefore$ total moles at new equilibrium

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$$n'_{total} = 2 \times 6 = 12$$

$$\underset{(g)}{PCl_5} \rightleftharpoons \underset{(g)}{PCl_3} + \underset{(g)}{Cl_2}$$

$$t = t, \quad \underset{2-x}{} \qquad \underset{2+x}{} \qquad \underset{y}{}$$

Where y = final amount of chlorine

total moles (new equilibrium)

$$(2-x)+(2+x)+y = 12$$

$$y = 8$$

Amount of Cl_2 at new equilibrium $= y =$ **8**

51. Final total moles = 12

Total pressure = 3 atm

$$P'_{PCl_5} = \frac{2-x}{12} \times 3 = \frac{2-x}{4}$$

$$P'_{PCl_3} = \frac{2+x}{12} \times 3 = \frac{2+x}{4}$$

$$P'_{Cl_2} = \frac{8}{12} \times 2 \text{ atm}$$

$$K_p = f(T)$$

[K_p is a function of temperature]

$$1 = \frac{\left(\frac{2+x}{4}\right) \times 2}{\left(\frac{2-x}{4}\right)}$$

$$3x = -2$$

$$x = -0.667$$

final amount of Cl_2 is 8 moles

final amount = initial amount + added amount + change from reaction shift

$$8 = 2 + (added) + x$$

$$8 = 2 + (added) - 0.667$$

$$\boxed{added = 6.667}$$

6.66 to 6.67

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